

MARMUL, OmanWMO: **413040**Lat: **18.1411N**Lon: **55.1806E**Elev: **273**StdP: **98.09**Time Zone: **4.00 (E04)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.6	12.2	-9.0	1.8	24.6	-6.2	2.3	24.4	12.0	24.2	10.7	23.2	3.2	220	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	44.0	20.2	43.1	20.4	42.2	20.5	25.2	29.8	24.8	29.9	24.3	30.5	4.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.4	20.0	26.8	23.7	19.2	26.6	23.0	18.4	26.5	79.4	29.6	77.4	29.4	75.3	30.1	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.4	9.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB	2.3	25.6	1.2	0.7	1.4	26.2	0.7	26.6	0.0	27.0	-0.9	27.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	19.6	21.4	25.2	29.0	32.3	34.6	35.0	33.9	30.7	27.7	24.2	20.9	(6)	
	DBStd	5.77	2.23	2.61	2.42	1.87	1.89	2.14	2.27	2.49	2.68	1.88	1.54	1.95	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	23	14	5	0	0	0	0	0	0	0	0	0	4	(9)	
	CDD10.0	6536	297	319	472	570	690	738	776	742	621	548	424	339	(10)	
	CDD18.3	3518	53	91	214	320	432	488	517	484	371	289	174	84	(11)	
	CDH23.3	47257	351	728	2117	3891	6201	7797	8367	7491	4899	3207	1612	597	(12)	
	CDH26.7	29612	35	192	991	2268	3995	5481	5971	5182	3086	1724	600	88	(13)	
(14) Wind	WSAvg	4.2	3.8	4.9	5.5	4.6	4.4	4.4	3.9	3.8	4.4	4.0	3.4	3.5	(14)	
(15) Precipitation	PrecAvg	93	8	7	12	10	6	6	6	3	6	6	10	13	(15)	
	PrecMax	197	17	24	89	84	46	60	26	39	19	35	107	43	(16)	
	PrecMin	25	3	0	1	0	0	0	1	1	1	1	2	1	(17)	
	PrecStd	45	4	6	20	18	10	11	6	8	4	7	19	10	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	32.4	37.0	40.0	43.6	45.2	44.9	44.2	42.9	38.8	33.9	29.6	(19)	
		MCWB	16.2	18.1	17.9	18.9	19.6	19.6	20.4	21.1	20.4	18.5	17.5	18.2	(20)	
	2%	DB	27.6	30.7	35.0	38.5	41.9	44.1	43.9	43.1	41.3	37.2	32.4	28.6	(21)	
		MCWB	16.9	17.5	18.1	18.6	19.2	19.5	20.5	21.2	19.8	17.9	17.7	18.0	(22)	
	5%	DB	26.4	29.1	33.3	37.4	40.8	43.0	43.0	42.1	40.0	35.6	31.1	27.7	(23)	
		MCWB	17.2	17.2	17.7	18.5	19.2	19.5	20.8	21.5	19.8	17.9	18.3	18.1	(24)	
	10%	DB	25.4	27.6	31.7	36.0	39.7	41.7	42.0	40.9	38.4	34.2	29.9	26.7	(25)	
		MCWB	16.8	17.2	17.5	18.5	19.4	19.7	21.0	21.5	19.8	18.1	18.3	17.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.8	22.4	23.2	24.6	26.1	25.7	24.8	24.6	24.0	24.0	23.2	21.9	(27)	
		MCDB	25.3	26.8	27.7	28.7	29.7	30.6	35.7	35.1	31.0	27.0	26.5	25.8	(28)	
	2%	WB	20.0	21.5	22.6	23.8	25.4	25.3	24.1	24.0	23.4	23.3	22.4	20.9	(29)	
		MCDB	23.6	25.6	26.9	27.6	29.1	30.2	35.2	35.6	29.9	26.5	25.9	24.9	(30)	
	5%	WB	19.2	20.6	22.1	23.3	25.0	24.7	23.7	23.6	22.9	22.8	21.7	20.2	(31)	
		MCDB	23.2	24.6	26.5	27.7	28.8	30.0	35.0	35.3	29.3	26.2	25.7	24.5	(32)	
	10%	WB	18.5	19.9	21.5	22.8	24.5	24.0	23.2	23.2	22.5	22.3	21.0	19.6	(33)	
		MCDB	22.8	24.0	26.0	27.7	28.7	30.4	35.0	34.9	28.7	26.0	25.4	24.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	12.8	13.4	14.2	14.4	14.1	13.9	13.9	14.3	13.0	12.5	12.0	(35)	
		MCDBR	12.4	13.8	15.3	15.7	15.8	15.1	14.4	14.2	15.6	15.0	13.4	12.4	(36)	
	5% WB	MCWBR	5.0	5.1	5.4	6.2	6.0	5.0	3.5	3.2	4.4	6.2	5.2	4.9	(37)	
		MCWBR	10.5	11.1	11.7	13.3	12.7	12.7	13.4	12.9	12.9	11.1	11.2	10.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.398	0.423	0.466	0.504	0.574	0.723	0.773	0.705	0.561	0.430	0.416	0.403	(40)	
		taud	2.105	2.021	1.897	1.828	1.688	1.443	1.405	1.512	1.738	2.045	2.085	2.091	(41)	
	Ebn at Noon		875	874	848	816	752	640	611	658	757	855	850	856	(42)	
		Edh at Noon	150	172	201	217	247	312	325	294	232	165	152	148	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.32	6.13	6.84	7.27	7.44	6.91	6.44	6.66	6.70	6.30	5.43	5.04	(44)	
	RadStd		0.12	0.11	0.22	0.19	0.16	0.29	0.30	0.17	0.12	0.14	0.16	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.15	N/A	-1.52	N/A	N/A	-0.30	N/A	N/A	N/A	N/A

Nomenclature: See separate page